

B.TECH.- THIRD SEMESTER
MID-SEMESTER EXAMINATION, SEPTEMBER 2025

Course Code-EEMTC04/ICMTC04
 Course Title- Mathematics III

Time: 1:30 hours

Max Marks-25

Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1(a)	Determine the constants a, b, c, d so that the function $f(z) = (x^2 + axy + by^2) + i(cx^2 + dxy + y^2)$ is analytic.	2.5	CO1
1(a)	Show that the transformation $w = \cos hz$ maps the lines parallel to x -axis and the lines parallel to y -axis into confocal central conics.	2.5	CO1
2(a)	Determine the poles and the residue at each pole of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$.	2.5	CO1
2(b)	Evaluate $\int_0^{2\pi} \frac{d\theta}{5 - 3 \cos \theta}$ by contour integration.	2.5	CO1
3(a)	Find the divergence and curl of the vector $\vec{F} = (x^2 + yz)\hat{i} + (y^2 + zx)\hat{j} + (z^2 + xy)\hat{k}$.	2.5	CO2
3(b)	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \vec{r} $, then show that $\text{grad } \phi(r) = \phi'(r) \frac{\vec{r}}{r}$.	2.5	CO2
4(a)	Find the directional derivative of the function $f(x, y, z) = 4xz^3 - 3x^2yz^2$ at $(2, -1, 2)$ along z -axis.	2.5	CO2
4(b)	Use Green's theorem to evaluate $\int_C (x^2y dx + x^2dy)$, where C is the boundary described counter clockwise of the triangle with vertices $(0, 0), (1, 0), (1, 1)$.	2.5	CO2

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5(a)	Show that $\int_C \vec{F} \cdot d\vec{r} = 3\pi$, where $\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$ and C is the arc of the curve $\vec{r} = \cos t \hat{i} + \sin t \hat{j} + t \hat{k}$ from $t = 0$ to $t = 2\pi$.	2.5	CO2
5(b)	Use divergence theorem to evaluate $\iiint_S \vec{A} \cdot \hat{n} dS$ where $\vec{A} = 2x^2y\hat{i} - y^2\hat{j} + 4xz^2\hat{k}$, taken over the region in the first octant bounded by $y^2 + z^2 = 9$ and $x = 2$.	2.5	CO2